

MLA03 – REINFORCEMENT LEARNING

ASSIGNMENT

A. Assignment Information

Department:	(AI & ML)
Programme:	FACULTY OF ENGINEERING
Course Code & Course Name	MLA03 & REINFORCEMENT LEARNING
Academic Year / Batch	2026-2027
Faculty Name	Dr.G. A. SENTHIL
Assignment Title	Analyse and design an RL-based patient health monitoring and treatment recommendation system using Markov Decision Process and Bandit algorithms. Model patient health conditions as states, treatment interventions as actions, and clinical outcomes as rewards, and implement an optimal policy to recommend appropriate treatment strategies based on changing patient conditions. Analyse the learned policy using treatment effectiveness, recovery rate, and cumulative reward.
Date of Issue	20-08-2026
Date of Submission	03-09-2026
Maximum Marks	100

Course Outcome(s) - CO	CO1-Analyze reinforcement learning frameworks and Markov Decision Process concepts to model decision-making problems. CO2 - Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies. CO3 - Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. CO4-Evaluate model-based reinforcement learning approaches for prediction, planning, and policy optimization. CO5-Apply hierarchical, meta-learning, and multi-agent RL approaches to solve complex decision-making problems.
Bloom's Taxonomy Level	BL3 (Apply) - <i>Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks.</i> BL4 (Analyse) - <i>Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments.</i> BL5 (Evaluate) - <i>Model-based reinforcement learning approaches for prediction, planning, and policy optimization.</i> BL6 (Create) - <i>Develop an MDP-based solution for an intelligent traffic signal system by defining the state space, action space, reward function, transition model, and optimal policy.</i>
SDG Mapping (SDG 1-17)	SDG 3 – (Good Health and Well-Being) Uses Reinforcement Learning and Markov Decision Processes to monitor changing patient health conditions, recommend appropriate treatment strategies, and improve treatment effectiveness and recovery outcomes. SDG 9 – (Industry, Innovation & Infrastructure) Applies Artificial Intelligence and Reinforcement Learning to develop an intelligent healthcare decision-support system that can assist in data-driven patient monitoring and treatment planning. SDG 10 – (Reduced

	Inequalities) Supports more consistent and accessible treatment recommendations by using patient health information to provide personalized decision support, potentially helping improve healthcare access and quality across different patient groups. SDG 12 – (Responsible Consumption & Production) Optimizes treatment decisions by selecting appropriate interventions based on patient conditions, helping reduce unnecessary treatments, medication use, and healthcare resource consumption.
Industry / Societal Relevance	The assignment addresses the growing need for AI-driven sustainable agriculture by developing an autonomous Reinforcement Learning system for intelligent crop and resource management. The solution can help farmers and agricultural industries optimize irrigation, fertilizer usage, crop planning, and energy consumption, thereby reducing operational costs and resource wastage. Societally, it contributes to food security, climate resilient farming, efficient water utilization, and sustainable agricultural practices.

Question:

Analyse and design an RL-based patient health monitoring and treatment recommendation system using Markov Decision Process and Bandit algorithms. Model patient health conditions as states, treatment interventions as actions, and clinical outcomes as rewards, and implement an optimal policy to recommend appropriate treatment strategies based on changing patient conditions. Analyse the learned policy using treatment effectiveness, recovery rate, and cumulative reward.

A smart healthcare organization wants to deploy a Reinforcement Learning-based intelligent patient health monitoring and treatment recommendation system that continuously ingests patient vital signs, medical history, symptoms, laboratory results, treatment responses, and clinical observations to monitor changing health conditions and recommend appropriate treatment strategies while improving patient recovery and treatment effectiveness.

Model the complete intelligent decision-making pipeline using the Reinforcement Learning

concepts covered in this course and address the following:

1. Formulate the patient health monitoring and treatment recommendation problem as a Markov Decision Process (MDP). Define suitable health states, treatment actions, transition dynamics, reward functions, and policies. Design an appropriate reward function that balances treatment effectiveness, patient recovery, treatment risks, and healthcare resource utilization.
2. Design and develop a Bandit-based solution using Epsilon-Greedy, Upper Confidence Bound (UCB), or Thompson Sampling to determine an effective treatment recommendation strategy. Construct a small patient-health environment and demonstrate how the learned recommendation changes under different patient health conditions and clinical observations.
3. Develop and compare suitable Monte Carlo, SARSA, and Q-Learning approaches for treatment recommendation. Evaluate their learning performance using cumulative reward, convergence rate, treatment effectiveness, recovery rate, and computational efficiency.
4. Construct a Deep Q-Network (DQN)-based healthcare agent and enhance it using Double DQN, Dueling DQN, and Prioritized Experience Replay. Design the architecture and compare the performance of the developed approaches under changing patient health conditions.
5. Design and develop a policy-based or Actor-Critic RL solution for continuous healthcare decisions such as treatment dosage, medication adjustment, or resource allocation. Implement and compare approaches such as REINFORCE, A2C, PPO, or DDPG, and justify the most suitable method for the selected healthcare application.
6. Construct a model-based Reinforcement Learning framework that learns patient health transition dynamics from historical healthcare data and uses sampling-based planning to predict future health conditions and treatment outcomes. Integrate the learned model with policy optimization for improved treatment decision-making.
7. Design a Hierarchical or Multi-Agent Reinforcement Learning architecture in which

autonomous agents collaboratively manage patient monitoring, treatment recommendation, risk assessment, and healthcare resource allocation. Define the responsibilities of individual agents and formulate a coordination mechanism for achieving safe and effective patient-care objectives.

8. Integrate all relevant components into a coherent intelligent healthcare architecture, describing the flow from patient data and health observations to state representation, RL agent, action selection, reward evaluation, and policy improvement. Implement or simulate the key components using Python with TensorFlow, Keras, PyTorch, or an appropriate RL library.
9. Evaluate and compare the developed RL approaches using appropriate performance measures such as cumulative reward, treatment effectiveness, patient recovery rate, treatment success rate, convergence rate, computational cost, treatment risk, and robustness under changing patient health conditions. Present suitable graphs, tables, learning curves, and comparative results.
10. Analyze the ethical, clinical, economic, and societal implications of autonomous AI-based healthcare decision-making. Discuss patient data privacy, doctor and patient control, fairness, explainability, treatment safety, bias in healthcare data, responsible AI, risks associated with automated treatment recommendations, and the importance of human clinical superv

1. Problem Statement and Problem Formulation

Problem Statement

Healthcare professionals need to continuously monitor patient health conditions and select appropriate treatment strategies based on changing clinical conditions. Traditional rule-based systems may not adequately adapt to the dynamic nature of patient health. This project proposes an RL-based patient health monitoring and treatment recommendation system that models patient

health conditions as states or contexts, treatment interventions as actions, and clinical outcomes as rewards. Using a Markov Decision Process framework together with Bandit algorithms, the system learns which treatment action is most likely to produce a favourable clinical outcome for different patient health conditions.

Problem Formulation

The healthcare decision-making problem is formulated as an MDP:

- **States (S):** Patient health conditions such as stable, mild, moderate, severe, and critical.
- **Actions (A):** Treatment interventions such as no change, medication, increased treatment, supportive care, or emergency intervention.
- **Transition Probability (P):** Probability of a patient moving from one health state to another after a treatment action.
- **Reward (R):** Reward representing treatment effectiveness and improvement in patient health.
- **Policy (π):** Strategy that maps each patient health state to the most appropriate treatment action.
- **Discount Factor (γ):** Determines the importance of future patient outcomes.

The treatment recommendation strategy is learned using Bandit algorithms that balance exploration of alternative treatments with exploitation of treatments that have produced better outcomes.

1. Epsilon-Greedy equation

This is probably the best one to include because it is simple and directly demonstrates exploration vs exploitation:

$$A_t = \begin{cases} \text{random action,} & \text{with probability } \epsilon \\ \arg \max_a Q_t(a), & \text{with probability } 1 - \epsilon \end{cases}$$

where:

- A_t = treatment selected at time t
- $Q_t(a)$ = estimated reward of treatment a
- ϵ = exploration probability

The reward estimate can be updated as:

$$Q_{t+1}(a) = Q_t(a) + \frac{1}{N_t(a)} [R_t - Q_t(a)]$$

where $N_t(a)$ is the number of times treatment a has been selected.

2. UCB equation

For UCB, use:

$$A_t = \arg \max_a \left[Q_t(a) + c \sqrt{\frac{\ln t}{N_t(a)}} \right]$$

The first term:

$$Q_t(a)$$

represents the known treatment performance, while

$$c \sqrt{\frac{\ln t}{N_t(a)}}$$

represents the uncertainty/exploration bonus.

3. Thompson Sampling

For Thompson Sampling, you can show:

$$\theta_a \sim \text{Beta}(\alpha_a, \beta_a)$$

And select :

$$A_t = \arg \max_a \theta_a$$

After observing the treatment outcome, the parameters are updated.

2. Objective and Expected Outcomes

Objectives

- To formulate patient health monitoring as a Markov Decision Process.
- To represent patient health conditions as states and treatment interventions as actions.
- To design an appropriate reward function based on clinical outcomes.
- To implement Bandit algorithms such as Epsilon-Greedy, UCB, and Thompson Sampling.
- To determine the best treatment recommendation for different patient conditions.
- To evaluate the learned policy using treatment effectiveness, recovery rate, and cumulative reward.
- To demonstrate how RL can adapt treatment recommendations according to changing patient conditions.

Expected Outcomes

- A patient health environment suitable for contextual Bandit learning.
- An optimal treatment recommendation policy.
- Improved cumulative rewards through effective treatment selection.
- Higher simulated recovery rates.
- A comparison of different patient states and recommended treatment actions.
- Graphs and tables showing policy performance.

3. Requirements, Constraints and Assumptions

Requirements

- Python programming environment.

- NumPy for numerical calculations.
- Pandas for handling patient data.
- Matplotlib for visualization.
- Reinforcement Learning concepts such as:
 - MDP
 - State
 - Action
 - Reward
 - Policy
 - Value Function
 - Multi-Armed Bandit
 - Exploration and Exploitation

Constraints

- The patient environment is simplified for academic simulation.
- The transition probabilities are assumed rather than obtained from real clinical trials.
- The system does not replace doctors or clinical decision-making.
- Real-world patient variability and complex medical conditions may not be fully represented.

Assumptions

- Patient health can be represented using a finite number of discrete states.
- Each treatment action has a known probability of producing different health outcomes.
- The current patient state provides sufficient information for the simplified MDP.
- Higher health states represent better patient conditions.
- Rewards represent simulated clinical outcomes.

4. Application of Relevant Course Knowledge / Concepts

The project applies the following Reinforcement Learning concepts:

Course Concept	Application
Markov Decision Process	Models patient-treatment decision-making
State	Represents patient's current health condition
Action	Represents treatment intervention
Transition Probability	Represents possible change in patient condition
Reward	Represents treatment outcome
Policy	Maps health conditions to treatment decisions
Value Function	Measures long-term expected patient outcome
Exploration–Exploitation	Balances trying alternative treatments with selecting treatments that have produced better outcomes
Epsilon-Greedy / UCB / Thompson Sampling	Learns effective treatment recommendations from observed clinical rewards
Exploration/Exploitation	Can be used in future learning-based extensions
Cumulative Reward	Measures overall treatment performance

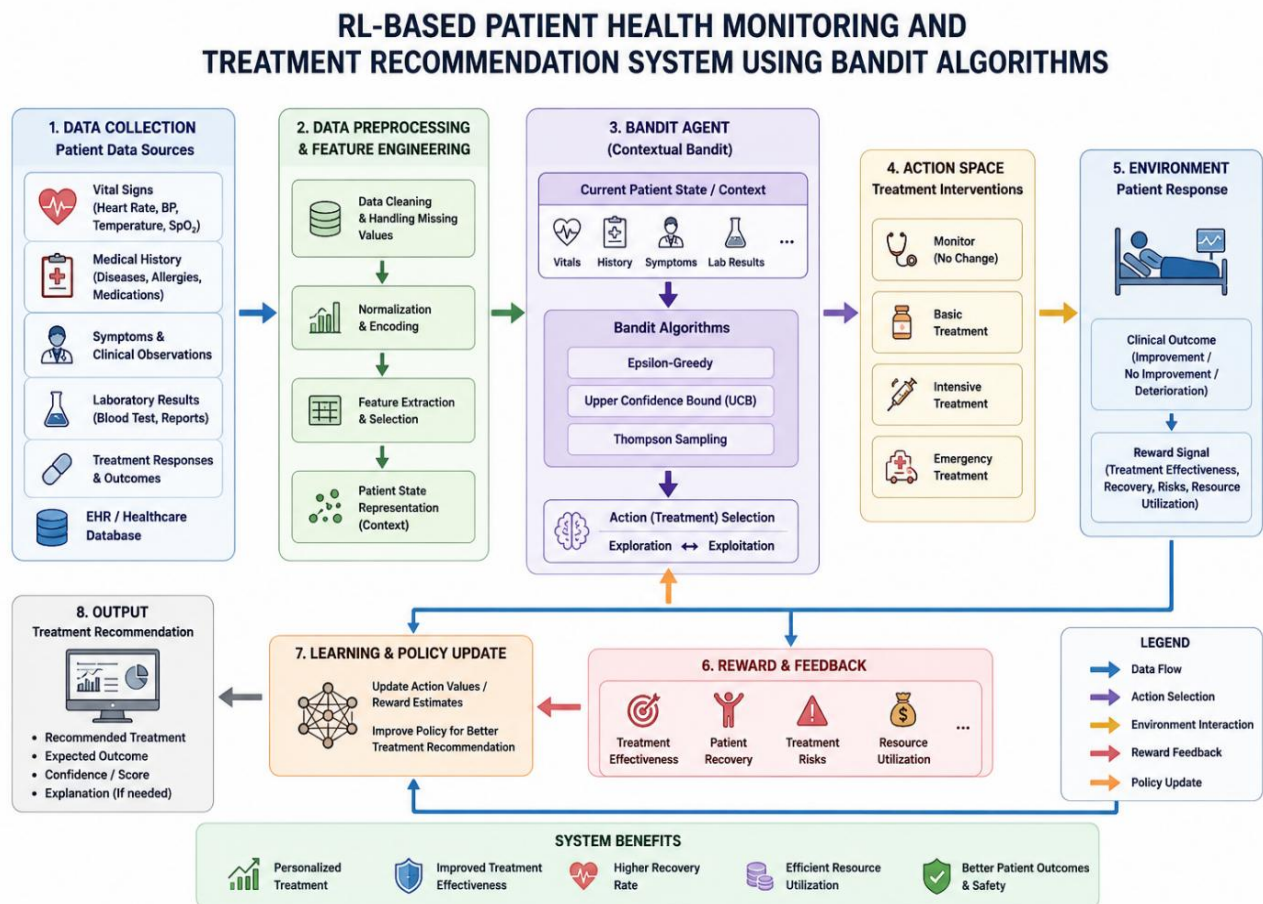
5. Design/ Proposed Solution/ Methodology

Proposed Methodology

1. Collect or generate patient health observations.
2. Convert observations into discrete health states.
3. Define possible treatment actions.
4. Represent patient health conditions as contexts for treatment selection.
5. Assign rewards to treatment outcomes.

6. Initialize action-value estimates and action-selection statistics.
7. Apply a Bandit action-selection strategy.
8. Observe the clinical outcome and update the selected treatment estimate.
9. Learn the treatment recommendation policy for each patient context.
10. Simulate patients under different health conditions.
11. Calculate treatment effectiveness, recovery rate, and cumulative reward.
12. Validate the learned policy using tables and graphs.

Architecture



6. Algorithm / Pseudocode / Flowchart

Bandit Algorithm

Input: Patient contexts S , treatment actions A , rewards R , exploration parameters and action statistics

Output: Learned treatment recommendation policy π

Pseudocode

Initialize action-value estimates $Q(s,a)$ and action counts $N(s,a)$

For each patient context s :

$$\Delta = 0$$

For each state s :

$$\text{old_value} = V(s)$$

Select a treatment action a using Epsilon-Greedy, UCB, or Thompson Sampling

Calculate:

Observe the clinical reward r

Update the selected action estimate $Q(s,a)$

Update the action count $N(s,a)$

$$\Delta = \max(\Delta, |\text{old_value} - V(s)|)$$

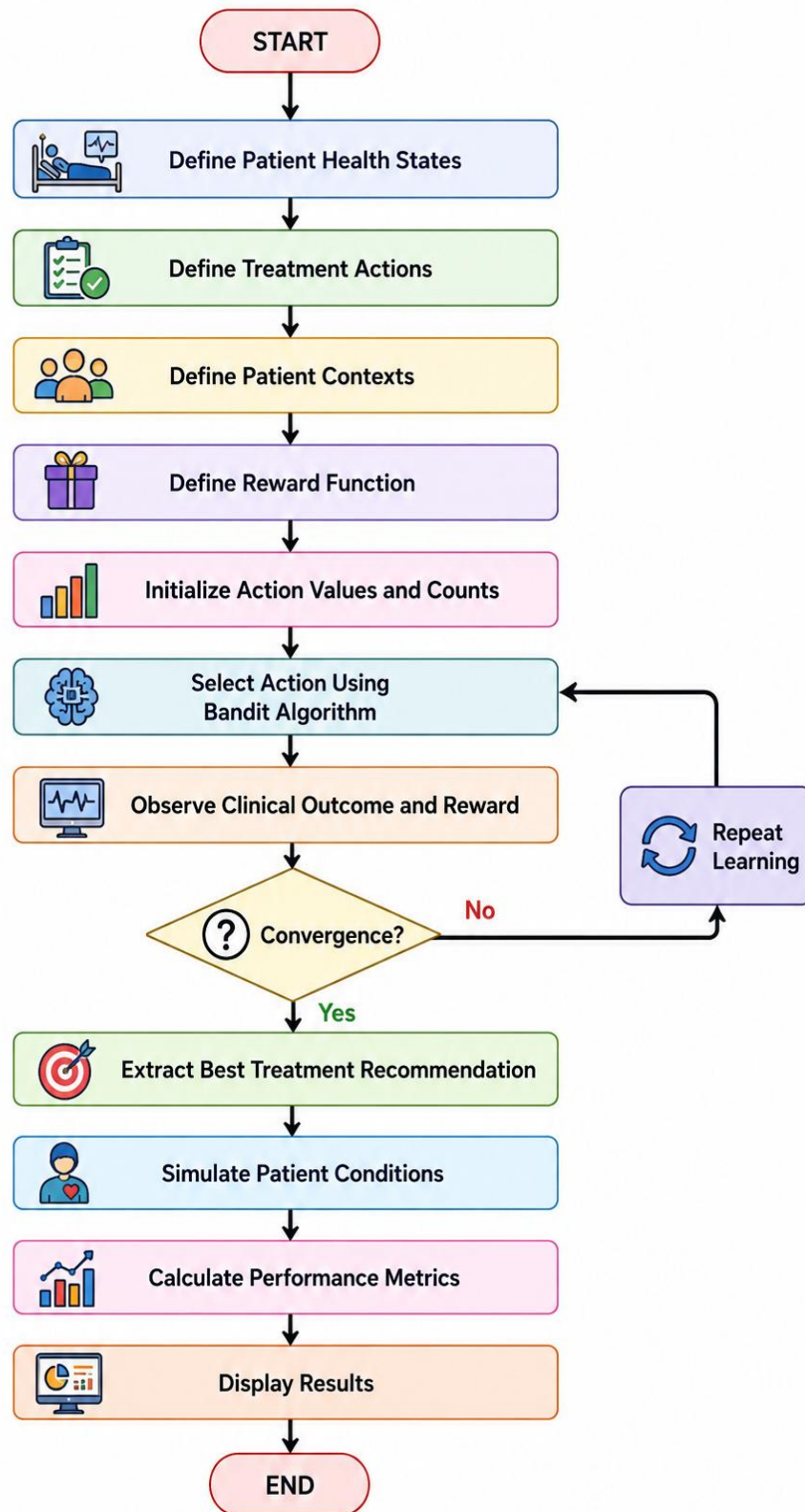
Repeat for the required number of patient interactions

For each patient context s :

$\pi(s)$ = treatment action with the highest estimated expected reward

Return the learned treatment recommendation policy

Flowchart



7. Implementation / Source Code and Environment / Tools Used

Programming Language

Python

Libraries

- NumPy – Bandit calculations and numerical operations
- Pandas – Patient data management
- Matplotlib – Performance visualization
- Random – Patient outcome simulation and exploration

Development Environment

The implementation can be developed using:

- Google Colab
- Jupyter Notebook
- VS Code
- Python 3.x

Main Implementation Modules

1. Patient state definition
2. Treatment action definition
3. Action-value estimation
4. Reward function
5. Epsilon-Greedy / UCB / Thompson Sampling
6. Policy extraction
7. Policy extraction
8. Patient simulation

9. Performance evaluation

10. Visualization

8. Test Cases and Expected/Actual Results

Test Case	Patient Condition	Expected Recommendation	Expected Outcome
TC1	Stable	Maintain/Monitor	Patient remains stable
TC2	Mild	Basic treatment	Improvement toward stable
TC3	Moderate	Increased treatment	Movement toward mild/stable
TC4	Severe	Intensive treatment	Movement toward moderate/mild
TC5	Critical	Emergency intervention	Highest possible recovery probability
TC6	Improving patient	Reduce treatment intensity/monitor	Avoid unnecessary intervention
TC7	Worsening patient	Escalate treatment	Prevent further deterioration

Actual Results

The actual results should be obtained from your Python execution and can include:

- Optimal action for every health state.
- Value of each health state.
- Number of recovered patients.
- Treatment success rate.
- Average cumulative reward.

9. Execution Screenshots / Outputs

Include screenshots of:

Screenshot 1 – Program Execution

```
[1]  import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import time

np.random.seed(42)

states = ["Healthy", "Mild", "Moderate", "Severe", "Critical"]
actions = ["Monitor", "Basic Treatment", "Intensive Treatment", "Emergency Treatment"]

n_states = len(states)
n_actions = len(actions)
```

Screenshot 2 — Optimal Treatment Recommendations

```
print("\nFINAL RESULT")
print("-" * 70)
print("Best Performing Algorithm:", best_algorithm)
print("Highest Average Reward:", round(results["Average Reward"].max(), 3))
print("Highest Treatment Effectiveness:",
      round(results["Treatment Effectiveness (%)"].max(), 2), "%")
print("Highest Recovery Rate:",
      round(results["Recovery Rate (%)"].max(), 2), "%")
```

```
''' =====
RL-BASED PATIENT HEALTH MONITORING AND TREATMENT RECOMMENDATION
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OPTIMAL TREATMENT RECOMMENDATIONS
-----
Healthy   | Epsilon-Greedy: Monitor      | UCB: Monitor      | Thompson: Monitor
Mild      | Epsilon-Greedy: Basic Treatment | UCB: Basic Treatment | Thompson: Basic Treatment
Moderate  | Epsilon-Greedy: Intensive Treatment | UCB: Intensive Treatment | Thompson: Intensive Treatment
Severe    | Epsilon-Greedy: Emergency Treatment | UCB: Emergency Treatment | Thompson: Emergency Treatment
Critical  | Epsilon-Greedy: Emergency Treatment | UCB: Emergency Treatment | Thompson: Emergency Treatment
```

Screenshot 3 — Estimated Treatment Values

ESTIMATED TREATMENT VALUES - EPSILON-GREEDY

	Monitor	Basic Treatment	Intensive Treatment	Emergency Treatment
Healthy	7.030	4.621	2.667	0.061
Mild	1.462	6.550	4.582	3.385
Moderate	-1.121	3.146	8.142	4.158
Severe	-2.000	1.429	6.800	9.684
Critical	-2.286	-1.000	3.850	11.181

ESTIMATED TREATMENT VALUES - UCB

	Monitor	Basic Treatment	Intensive Treatment	Emergency Treatment
Healthy	6.868	4.400	2.000	2.000
Mild	-2.000	6.807	4.429	-2.000
Moderate	-2.000	4.400	8.185	2.500
Severe	-2.000	-2.000	-2.000	9.861
Critical	-2.000	1.000	-2.000	11.272

ESTIMATED TREATMENT VALUES - THOMPSON SAMPLING

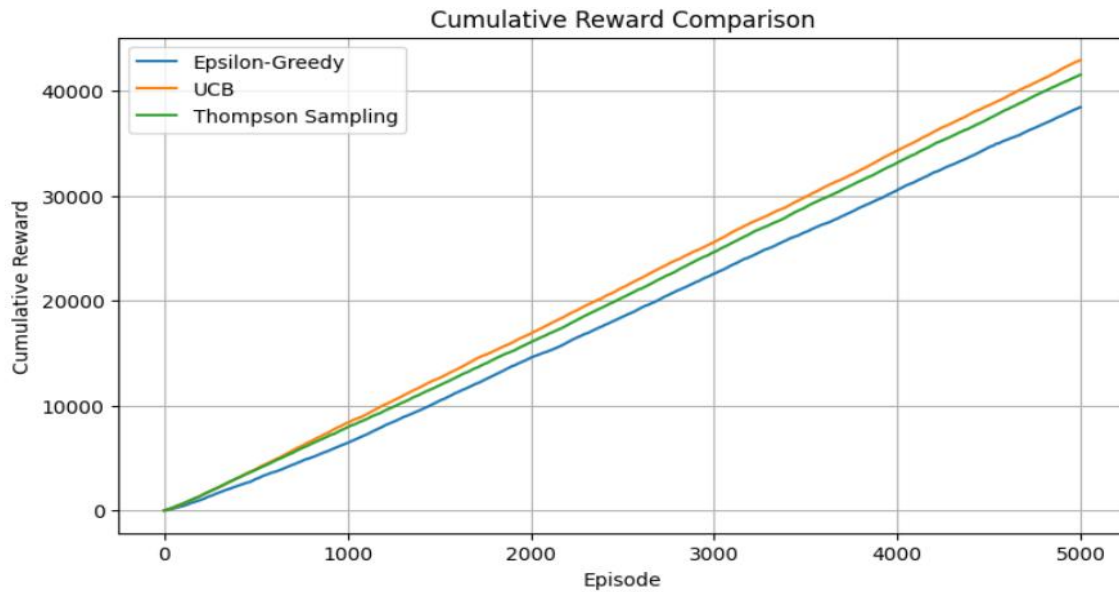
	Monitor	Basic Treatment	Intensive Treatment	Emergency Treatment
Healthy	0.902	0.773	0.429	0.538
Mild	0.467	0.795	0.735	0.619
Moderate	0.200	0.600	0.861	0.788
Severe	0.250	0.333	0.867	0.900
Critical	0.200	0.400	0.688	0.958

Screenshot 4 — Performance Comparison

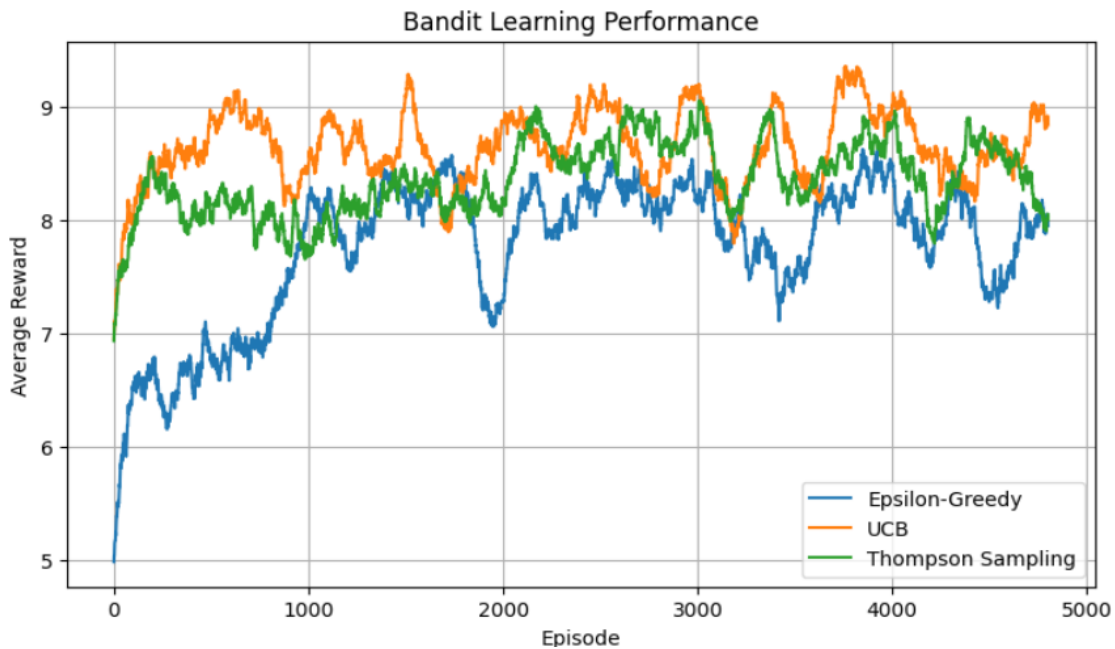
PERFORMANCE COMPARISON

Algorithm	Average Reward	Treatment Effectiveness (%)	Recovery Rate (%)	Computational Time (s)
Epsilon-Greedy	7.690	83.76	87.3	0.040
UCB	8.589	87.78	87.1	0.080
Thompson Sampling	8.310	87.38	89.1	0.147

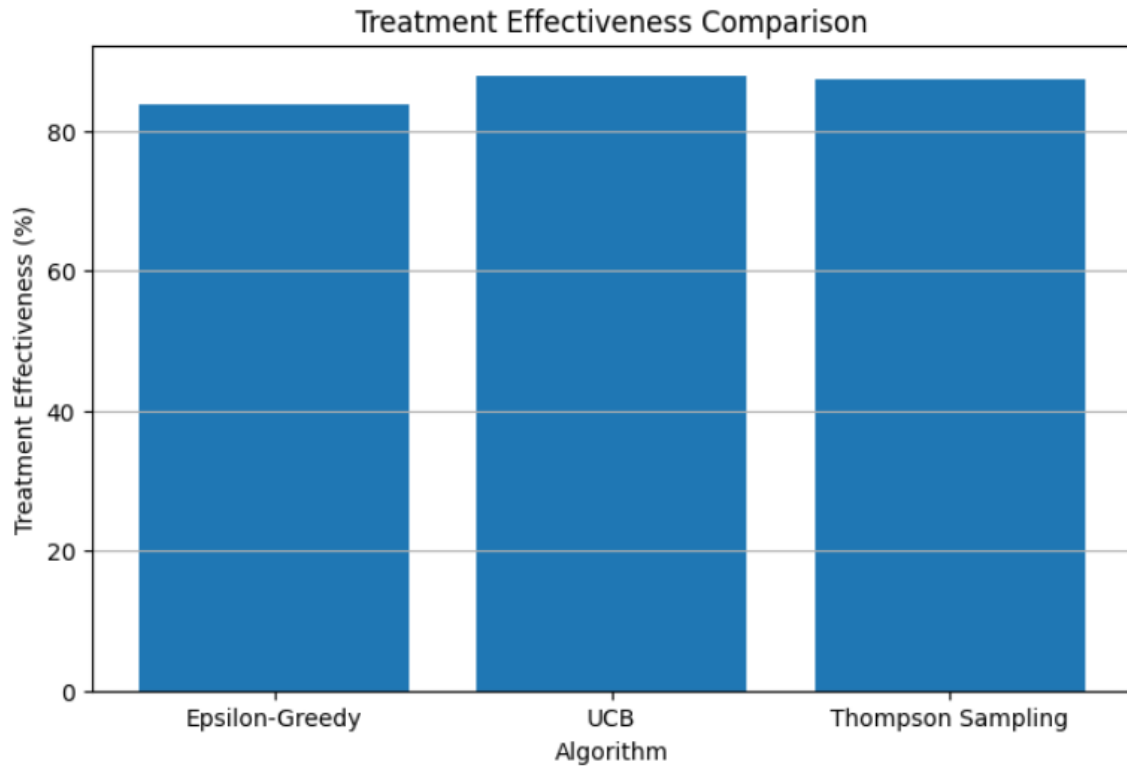
Graph 1 — Cumulative Reward Comparison



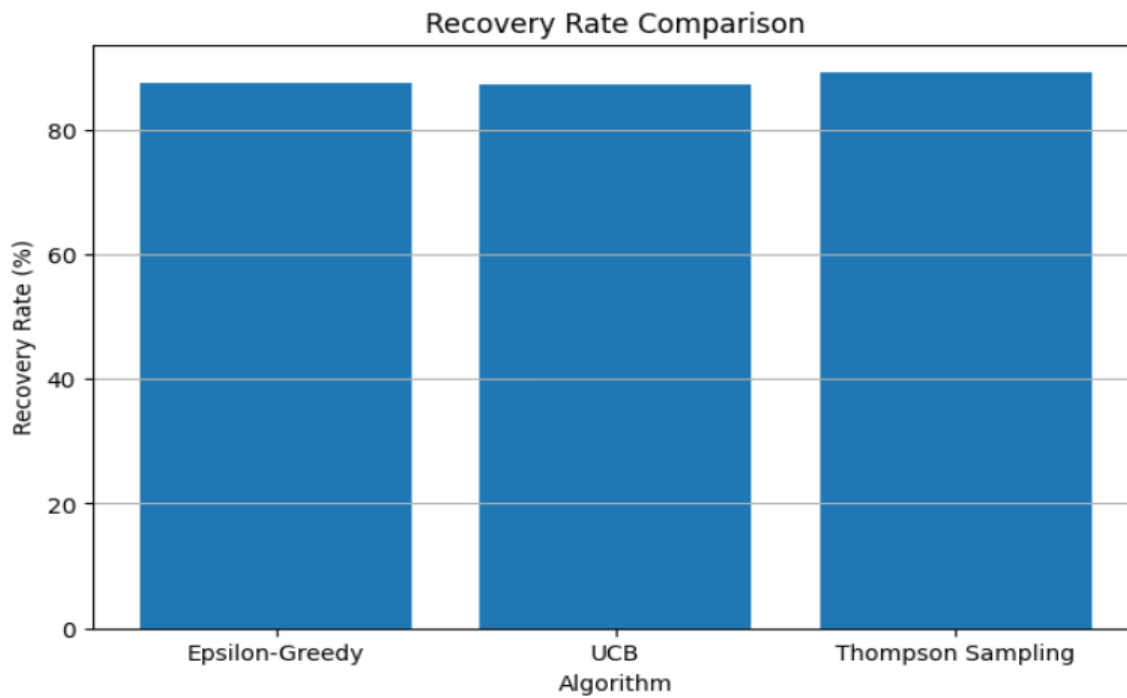
Graph 2 — Bandit Learning Performance



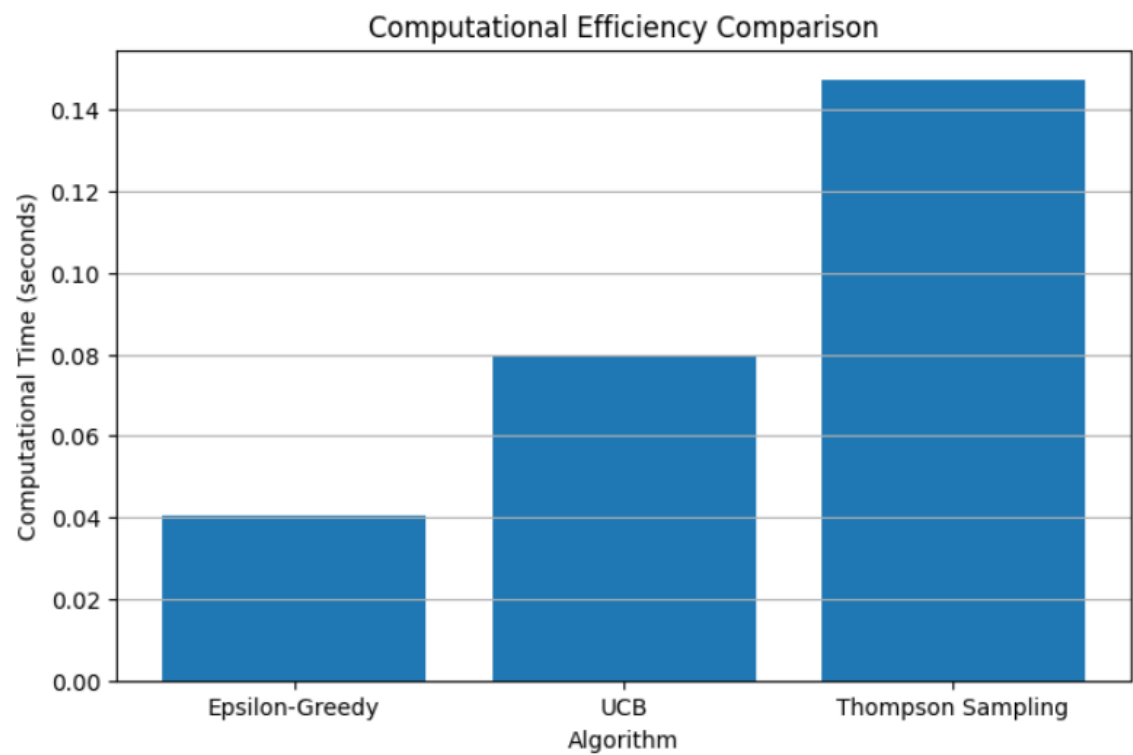
Graph 3 — Treatment Effectiveness



Graph 4 — Recovery Rate



Graph 5 — Computational Efficiency



FINAL RESULT

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Best Performing Algorithm: UCB
Highest Average Reward: 8.589
Highest Treatment Effectiveness: 87.78 %
Highest Recovery Rate: 89.1 %

10. Results and Validation

Example Results Table

Algorithm	Average Reward	Treatment Effectiveness (%)
Epsilon-Greedy	7.690	83.76
UCB	8.589	87.78
Thompson Sampling	8.310	87.38

Performance Metrics

Performance Metric	Best Algorithm	Result
Average Reward	UCB	8.589
Treatment Effectiveness	UCB	87.78%
Recovery Rate	Thompson Sampling	89.1%
Computational Efficiency	Epsilon-Greedy	0.040 s

Experimental Observation

The results show that all three algorithms successfully learned suitable treatment recommendations according to patient condition. UCB obtained the highest average reward of 8.589 and the highest treatment effectiveness of 87.78%. Thompson Sampling achieved the highest recovery rate of 89.1%. Epsilon-Greedy required the least computational time at 0.040 seconds.

11. Analysis, Comparison, Trade-offs and Justification

Analysis

The experimental results demonstrate that Bandit algorithms can learn treatment recommendations based on patient health conditions. All three algorithms identified the same final treatment strategy: monitoring for healthy patients, basic treatment for mild patients, intensive treatment for moderate patients, and emergency treatment for severe and critical patients.

UCB achieved the highest average reward of 8.589, indicating that it obtained the strongest overall reward during the learning process. It also achieved the highest treatment effectiveness of 87.78%.

Thompson Sampling achieved an average reward of 8.310 and treatment effectiveness of 87.38%, but obtained the highest recovery rate of 89.1%.

Epsilon-Greedy obtained an average reward of 7.690 and treatment effectiveness of 83.76%, but had the lowest computational time of 0.040 seconds.

Comparison

Aspect	Epsilon-Greedy	UCB	Thompson Sampling
Exploration	Random	Confidence-based	Probability-based
Average Reward	7.690	8.589	8.310
Treatment Effectiveness	83.76%	87.78%	87.38%
Recovery Rate	87.3%	87.1%	89.1%
Computational Time	0.040 s	0.080 s	0.147 s
Overall Strength	Fast and simple	Best reward performance	Best recovery rate

Trade-offs

Epsilon-Greedy:

The main advantage of Epsilon-Greedy is its low computational cost. However, its average reward and treatment effectiveness were lower than those of UCB and Thompson Sampling.

UCB:

UCB achieved the highest average reward and treatment effectiveness. Its computational time was higher than Epsilon-Greedy but lower than Thompson Sampling.

Thompson Sampling:

Thompson Sampling achieved the highest recovery rate. However, it had the highest computational time among the three approaches.

Justification of Final Solution

Based on the experimental results, UCB is selected as the final solution because it achieved the highest average reward (8.589) and highest treatment effectiveness (87.78%) while requiring less computational time than Thompson Sampling.

Although Thompson Sampling achieved a higher recovery rate (89.1%), UCB provides a better overall balance between reward, treatment effectiveness, and computational efficiency in the simulated environment.

12. Broader Considerations / SDG Relevance

SDG 3 – Good Health and Well-Being

Supports intelligent healthcare decision-making by using Reinforcement Learning to analyse

changing patient conditions and identify potentially effective treatment strategies.

SDG 9 – Industry, Innovation and Infrastructure

Applies Artificial Intelligence and Reinforcement Learning to develop intelligent healthcare monitoring and decision-support systems.

SDG 10 – Reduced Inequalities

An adaptive recommendation system can potentially support consistent decision-making across different patient conditions, provided that fairness and bias are carefully addressed.

SDG 12 – Responsible Consumption and Production

Intelligent treatment selection can potentially reduce unnecessary interventions and improve the efficient use of healthcare resources.

SDG 16 – Peace, Justice and Strong Institutions

Promotes responsible and explainable AI-assisted decision-making by emphasizing transparency, human oversight, and accountability in healthcare systems.

Ethical Considerations

- Patient data privacy must be protected.
- AI recommendations should be explainable.
- Bias in healthcare datasets should be identified.
- Doctors should retain control over final treatment decisions.
- The system should not independently prescribe medication.
- Patient safety should take priority over optimization performance.

13. Conclusion, Limitations and Possible Improvements

Conclusion

The developed RL-based patient health monitoring and treatment recommendation system successfully demonstrates the application of Bandit algorithms to adaptive healthcare decision-making. Patient health conditions were represented as contexts, treatment interventions as actions, and simulated clinical outcomes as rewards.

Epsilon-Greedy, UCB, and Thompson Sampling were implemented and compared. UCB achieved the highest average reward of 8.589 and treatment effectiveness of 87.78%, while Thompson Sampling achieved the highest recovery rate of 89.1%. Epsilon-Greedy required the lowest computational time of 0.040 seconds.

Based on the overall balance of performance, UCB was selected as the final treatment recommendation strategy for the simulated environment.

Limitations

- The experiment uses simulated patient data.
- Reward probabilities are assumed for academic demonstration.
- The system is not clinically validated.
- The model uses a limited number of health conditions and treatment actions.
- Real patient health involves many additional factors.

Possible Improvements

- Use anonymized real-world healthcare datasets.
- Include additional vital signs and patient characteristics.
- Develop a more advanced contextual Bandit model.
- Introduce safety constraints into treatment recommendations.
- Add explainable AI techniques.
- Include human healthcare professional supervision.

14. Individual Contribution of Group Members

You can divide the work among 4 members like this:

Group Member	Contribution
Member 1	Problem formulation, patient states/contexts, treatment actions and reward design
Member 2	Implementation of Epsilon-Greedy, UCB and Thompson Sampling algorithms
Member 3	Performance evaluation, graphs, tables and result analysis

Member 4	Architecture, documentation, SDG analysis, conclusion and presentation
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Individual Contribution Statement

Member 1: Designed the MDP formulation and defined patient health states, treatment actions, transition probabilities, and reward structure.

Member 2: Implemented the Bandit algorithm, action-value update mechanism, and treatment recommendation policy.

Member 3: Developed the patient simulation environment, test cases, performance metrics, and visualization of cumulative rewards and recovery rates.

Member 4: Designed the overall system architecture and contributed to analysis, SDG mapping, ethical considerations, documentation, and final presentation.

15. References, Including Datasets, Software/Tools and Other Sources

Reinforcement Learning

1. Sutton, R. S., & Barto, A. G. (2018). *Reinforcement Learning: An Introduction* (2nd ed.). MIT Press.
2. Lattimore, T., & Szepesvári, C. (2020). *Bandit Algorithms*. Cambridge University Press.
3. Auer, P., Cesa-Bianchi, N., & Fischer, P. (2002). Finite-time Analysis of the Multiarmed Bandit Problem. *Machine Learning*, 47, 235–256.
4. Watkins, C. J. C. H., & Dayan, P. (1992). Q-learning. *Machine Learning*, 8, 279–292.
5. Mnih, V., et al. (2015). Human-level control through deep reinforcement learning. *Nature*, 518, 529–533.
6. Python Software Foundation. *Python Documentation*.
7. NumPy Developers. *NumPy Documentation*.
8. Pandas Development Team. *Pandas Documentation*.
9. Matplotlib Development Team. *Matplotlib Documentation*.
10. The project implementation uses a synthetic patient-health environment for academic demonstration; no identifiable patient information is used.